



Anatomically Prior-Enhanced X-Ray Network for Thoracic Disease Classification and Localisation

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Project Overview

APEX-Net is an AI-powered system that analyses chest X-ray images to automatically detect different thoracic diseases and highlight the affected areas.



**Assist Doctors in
Underserved Areas**

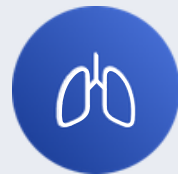


**Catch Rare Diseases
Early**



**Make AI Decisions
Transparent**

Why Does This Problem Matter?



Global Threat

Chest diseases affect **hundreds of millions** of people worldwide every year.



Most Common Test

Chest X-rays are the **cheapest and most widely used** diagnostic tool available.



Shortage of Specialists

Many regions, particularly in low-income countries, **lack enough trained radiologists**.



Clinical Need

An automated system can help doctors make **faster, more reliable** decisions.

Three Critical Gaps in Existing AI Systems

1

**Rare Disease
Blindness**

2

No Anatomical Focus

3

No Visual Explanation

APEX-Net: One System, Three Answers

Balanced Learning

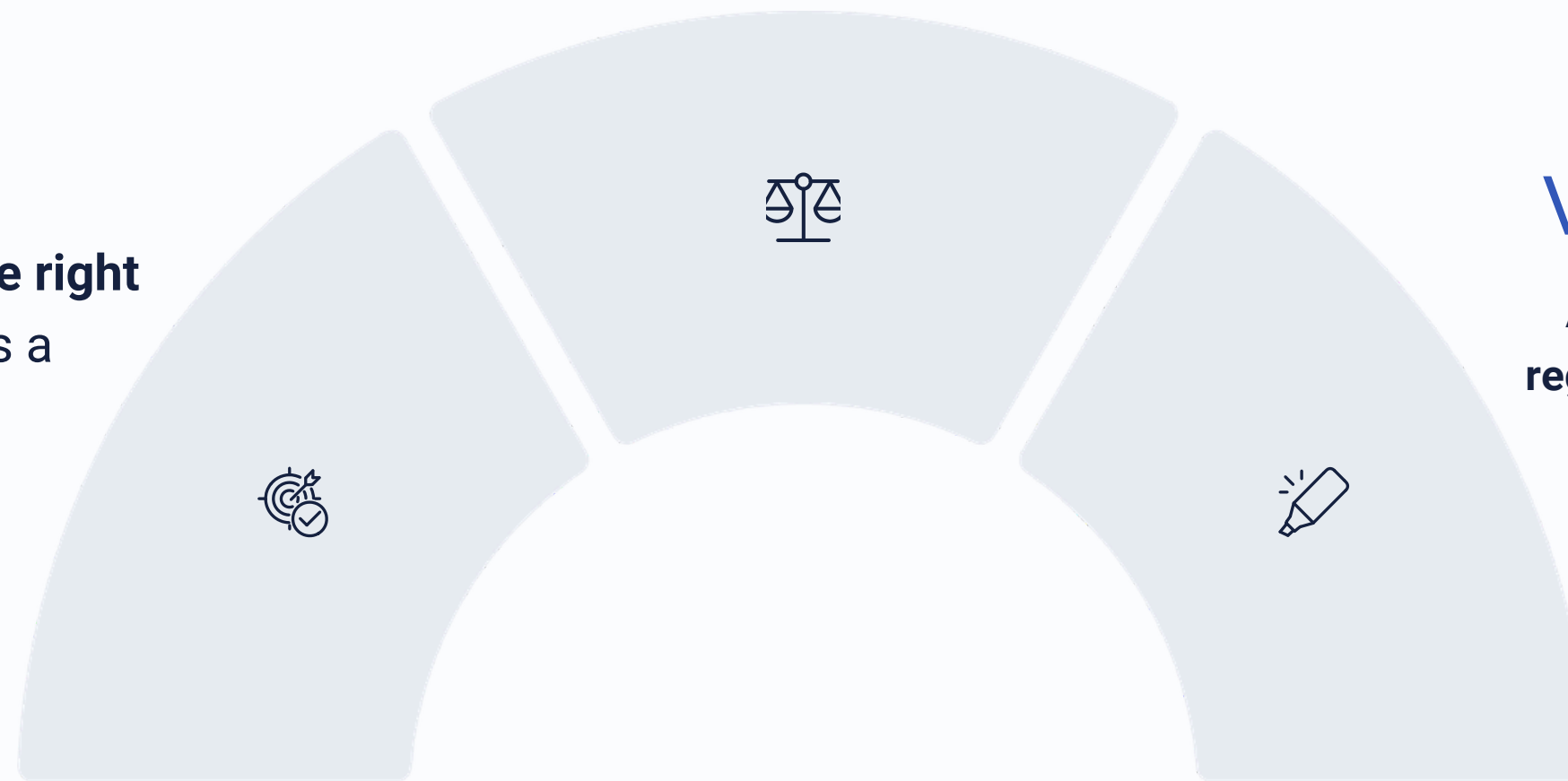
A specialised training strategy that gives **extra emphasis to rare diseases**, reducing missed diagnoses.

Smart Attention

Guides the model to focus on **the right anatomical regions** — just as a radiologist would.

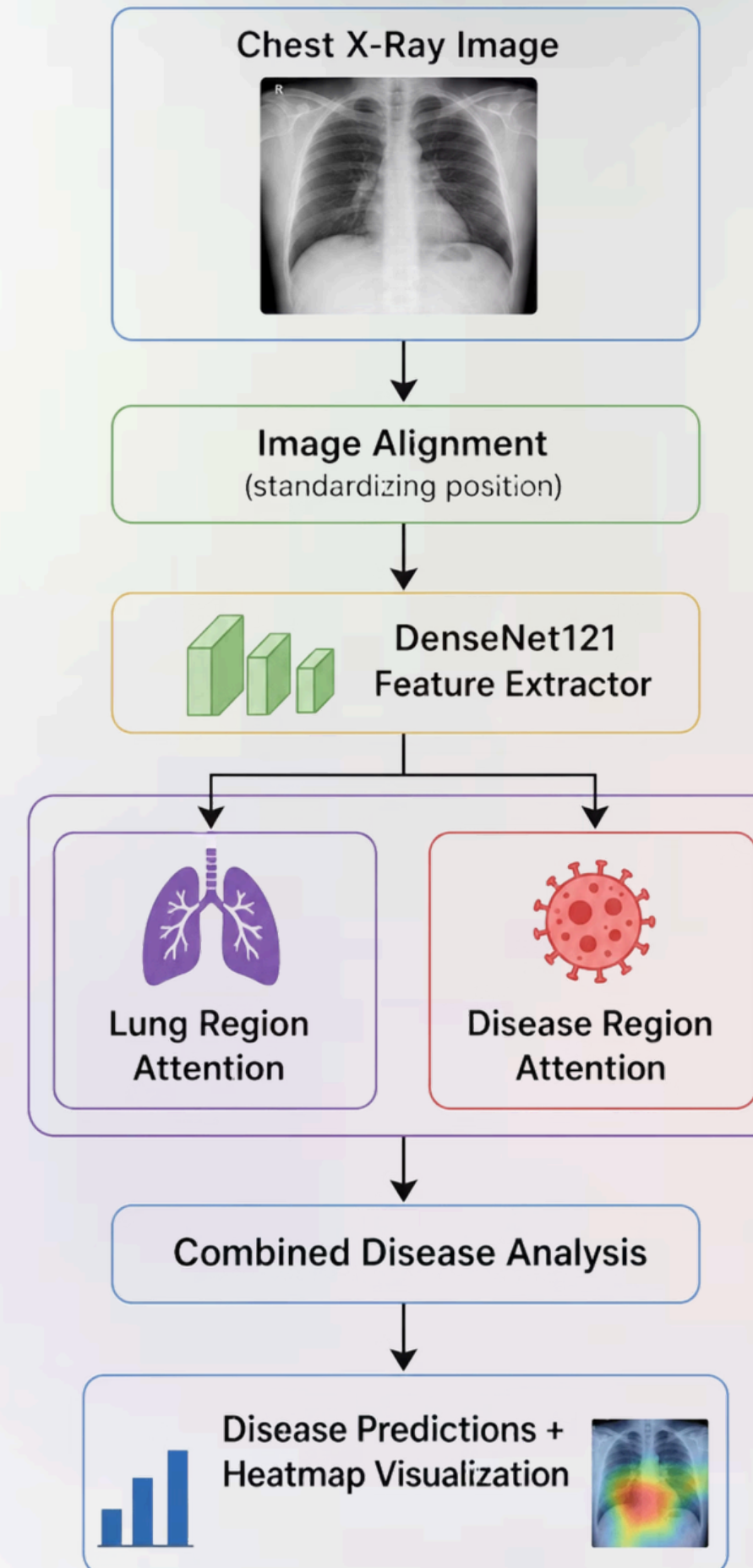
Visual Explanation

Automatically **highlights the affected region** in the X-ray so clinicians can verify every prediction.



METHODOLOGY

How APEX-Net Works



EXPECTED OUTCOMES

What This Project Will Deliver



Disease Detection

Accurately classify **14 distinct chest diseases** from a single X-ray image.



Localisation

Highlight **exactly where** in the X-ray each detected disease is located.



Web Application

A working **demonstration interface** that clinicians or reviewers can interact with directly.

Thank You